

DIPANGI AGARWAL

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SUMMARY

Full-Stack Developer skilled in Python, FastAPI, JavaScript and PostgreSQL, building scalable REST API-driven web applications development — from database schema to responsive frontend — with a focus on performance and measurable business impact.

SKILLS

- **Languages & Frontend** — Python, HTML, CSS, JavaScript
- **Backend & Databases** — FastAPI, REST APIs, SQLAlchemy, Celery, Redis, Socket.io, MySQL, PostgreSQL, MongoDB, Supabase
- **Tools & Platforms** — Git & GitHub, Postman, Docker, Linux (Ubuntu 22.04), Render, Vercel, JWT
- **Libraries/ Frameworks** — Numpy, Pandas, Matplotlib, Seaborn, Scikit-learn, NLTK, OpenCV
- **Core Concepts** — OOP, Data Structures & Algorithms (Basics), Debugging & Troubleshooting, Code Reviews

EXPERIENCE

Full Stack Developer Intern, Pixel Genix IT Solutions 01/2026 – 06/2026 | Jaipur, India

Project: Pixel CRM — A Multi-Tenant SaaS Web Application used by multiple client organizations to manage leads, deals, and subscription billing in one platform.

- Designed, developed, tested, and maintained **30+ RESTful API endpoints and backend services** (Python, FastAPI, SQLAlchemy, PostgreSQL) for a multi-tenant platform; Postman-based testing and founder-led **code reviews** cut post-release bugs by an estimated **30%** compared to the prior release cycle.
- Automated organization onboarding, bulk user invites, and CSV export workflows, cutting manual admin setup time per new organization from **2 hours to under 1 hour**.
- Migrated transactional email from Gmail SMTP to Brevo HTTPS API, **cutting email delivery failures by 95%** and unblocking deployment on Render's free tier.
- Implemented Redis-backed rate limiting, JWT auth, and forgot/reset password flows; **blocked all simulated brute-force login attempts during QA testing**.
- Debugged deployment and data-migration issues across Render, Celery/Redis, and Supabase, **cutting subscription-reminder task time by 35%** via Celery Beat scheduling, while collaborating via **Git/GitHub** for version control and code reviews toward production readiness.

Project: AI-Powered Attendance Management System — An internal tool built to replace manual attendance registers for a 100+ employee organization using face recognition.

- Developed and deployed a **full-stack web app** (FastAPI + React) for 100+ employees, **cutting attendance-marking time to under 30 seconds per session**.
- Configured **Tailscale** to securely connect the on-site camera's local Wi-Fi network to the Linux VPS, enabling reliable remote video feed access without public port exposure.
- Built a **responsive React frontend** with sidebar navigation, cookie-based authentication, and multi-select batch management, **cutting admin batch-assignment time by roughly half**.
- Migrated face recognition from **face_recognition** library to **InsightFace's buffalo_l** model (512-dim embeddings), **achieving 94% real-time recognition accuracy** across **45+ tested REST API endpoints (Postman manual unit testing)**.
- Implemented Celery/Redis background processing and **Docker** containerization, **reducing deployment setup time by an estimated 25%**; automated WhatsApp absence alerts via WAHA API (WhatsApp HTTP API) and APScheduler, **eliminating manual daily attendance reporting**.

EDUCATION

B. Tech in Computer Science and Engineering (AI&ML Specialization), JECRC University 09/2022 – 07/2026 | Jaipur, India
CGPA: 8.77/10

PROJECTS

AI-Powered Video Analysis System — RAG-Based Video Q&A Chatbot

- Built an end-to-end pipeline to extract audio from video using **FFmpeg** and transcribe it (speech-to-text) with **OpenAI Whisper**.
- Implemented text chunking and embedding generation with the **Ollama bge-m3 model**, enabling semantic retrieval.
- Built a **Retrieval-Augmented Generation (RAG)** chatbot using the **llama3.2 model**, letting users ask natural-language questions and get accurate answers sourced directly from video content.

CERTIFICATIONS

- IBM SkillBuild — "Python For Data Science" Certificate [Sep 2025]
- Samatrix Consulting Pvt. Ltd. — ML Lab (Python, Scikit-learn, TensorFlow), Stats, Foundation in AI/Data Science [2023-2025]
- Internship Studio — AI Training & Common Internship Test Participation Certificate [Jun 2024-Jul 2024]